Group 21 | 11/02/26

Open Octave

A modular, robotically-enhanced educational keyboard system

Our Product

Early-stage piano education, reimagined.

- Modular Design
- Unique Robotic Learning Modes
- Teacher-Facing Software Application

Target Audience

BGE Phase Learning
Likely within the Second to Third CfE levels

Designed For Complete Beginners
Most utility for those who have never touched a keyboard

Music Technology Market

Music EdTech workshop formats are poised for a **16.3% CAGR to 2031.** (2)

The UK government's Industrial Strategy: **double business investment** in the creative industries, from £17 billion to £31 billion by 2035 (3)

The UK creative tech sector attracted the **third highest amount** of foreign venture capital investment behind the US and China in 2021 (4)

The Existing Disparity

£1,865

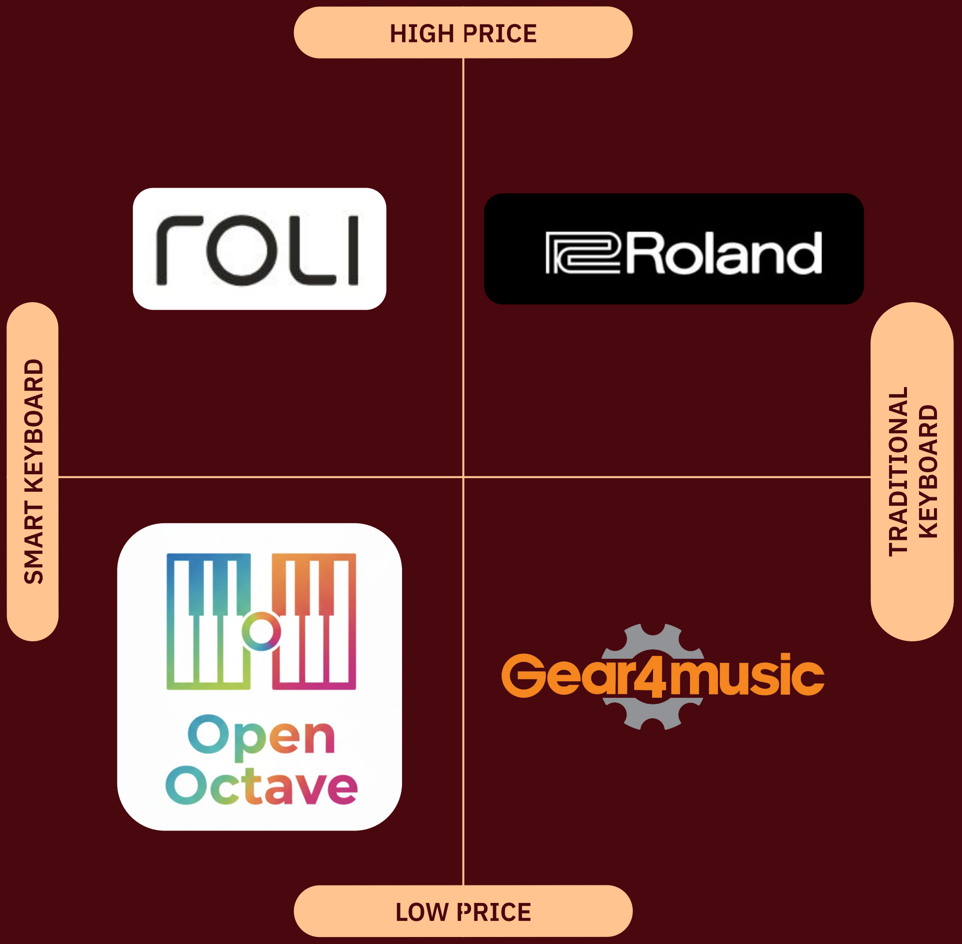
The mean yearly budget for music departments in **state schools**

£9,917

The mean yearly budget for music departments in **private schools**

(5)

Our Competitors



- Smart learning vs. traditional learning
- Budget-friendly at scale
- Classroom vs. home

Primary Goals for Demo 1

Hardware →

- Functional and responsive key mechanism consisting of at least **two** working keys
- Functional robotic auto-pressing and lighting up of keys

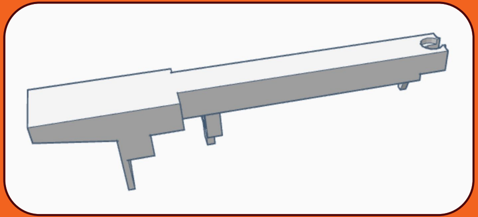
Firmware →

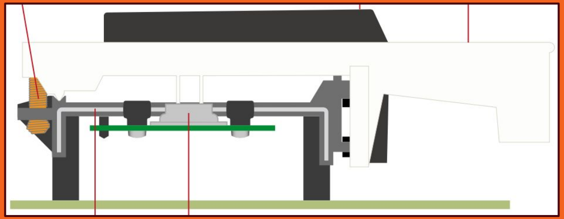
- Simple program to enable and control the key mechanisms

Mechanical Engineering

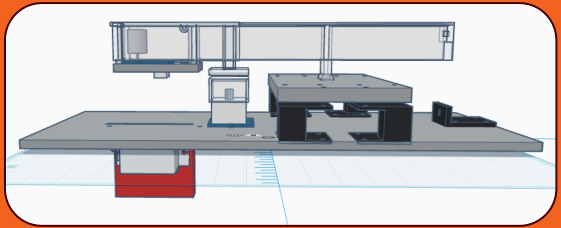
Our attempted methods to build keys that fit the Open Octave criteria

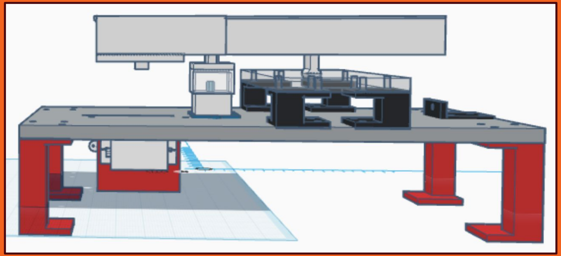
1. Researching pre-existing 3D models



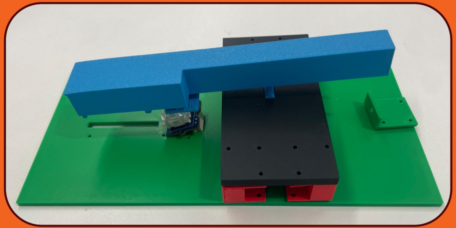


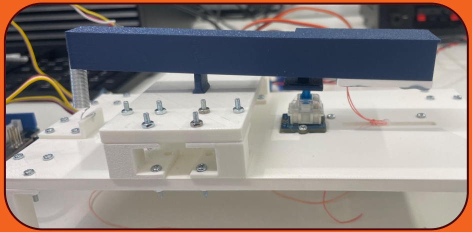
2. Designing our bespoke solution





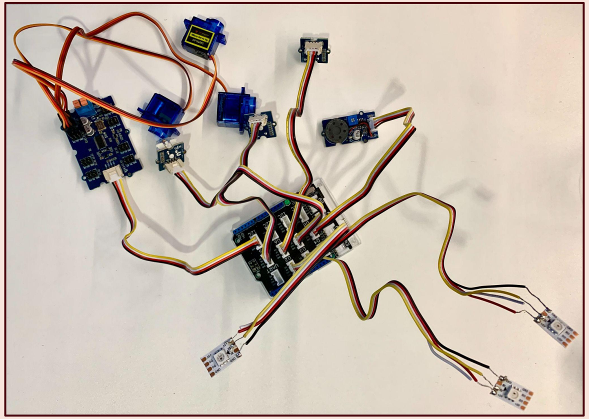
3. Iterative manufacturing

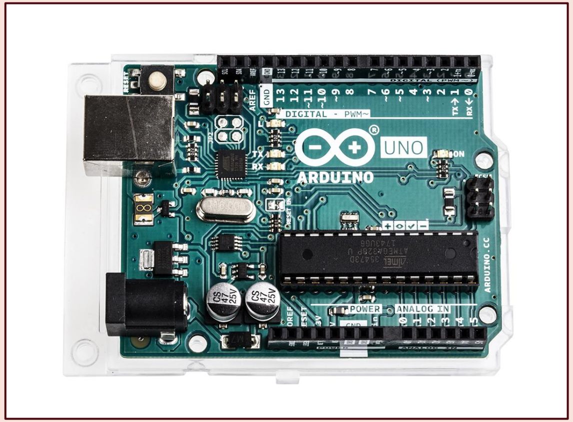




Electrical Engineering

The steps we took to turn our physical mechanisms into a working system






```
========================================
         SERIAL COMMANDS
========================================
MODE:
  m - Switch to MANUAL mode
  a - Switch to AUTOMATIC_LEDS mode
  f - Switch to FULL_AUTOMATIC mode
SEQUENCE:
  s - Start sequence
  x - Stop sequence
  n - Next sequence
  p - Previous sequence
  0-9 - Select sequence by number
  l - List all sequences
TESTING:
  g - Toggle test log mode (CSV output)
  t - Test LEDs
  u - Test servos
HELP:
  h - Show this help
========================================
```

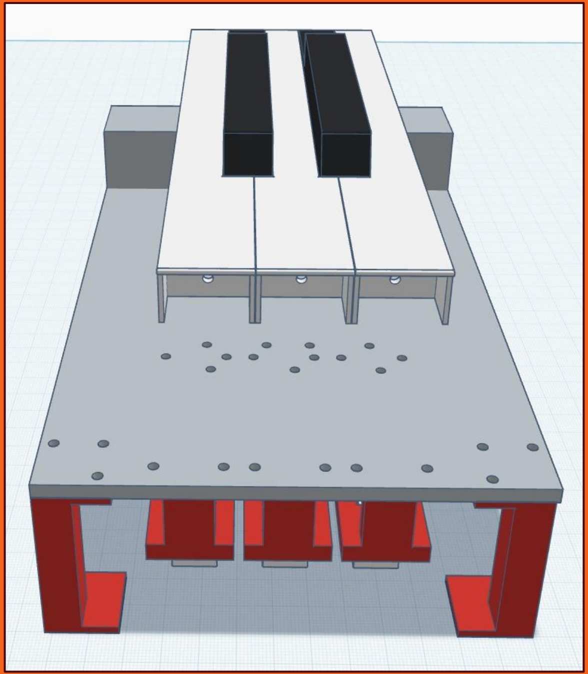
1. Component Selection

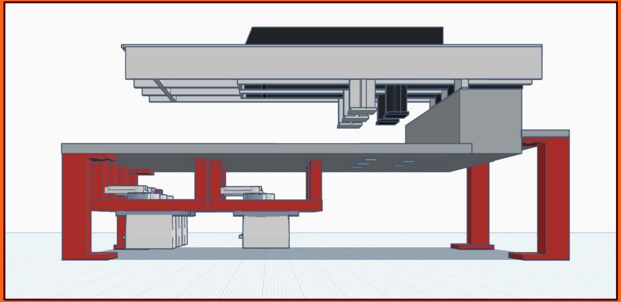
2. Embedded Programming

3. Serial Interface

What’s next for Open Octave?

Demo 2 and Beyond...





Mechanics:

Extending the current hardware, designing for modularity

Electronics:

Exposing a useful API, designing for networking

Software:

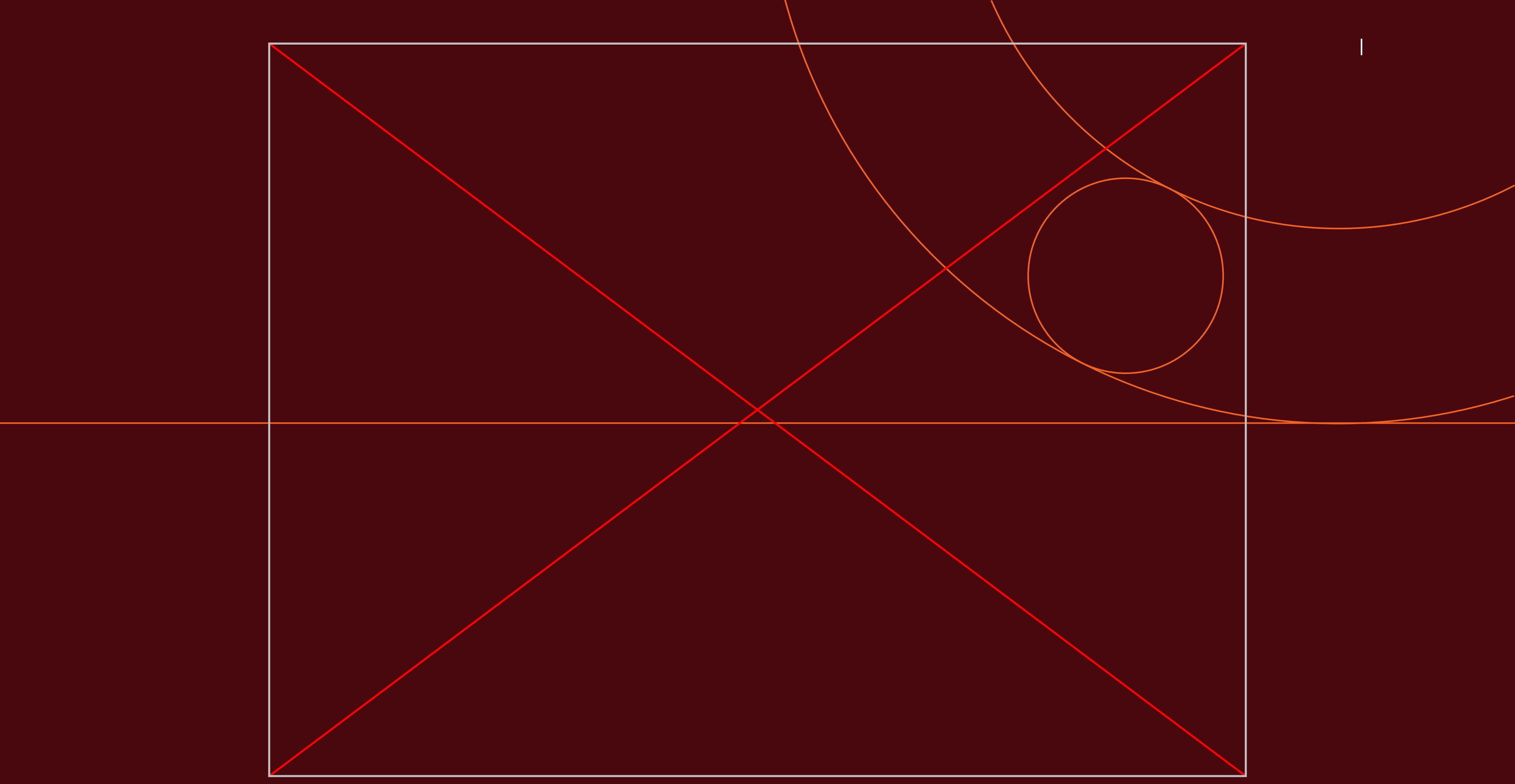
Teacher-facing application, MIDI processing

Open Octave

Thank You!

References:

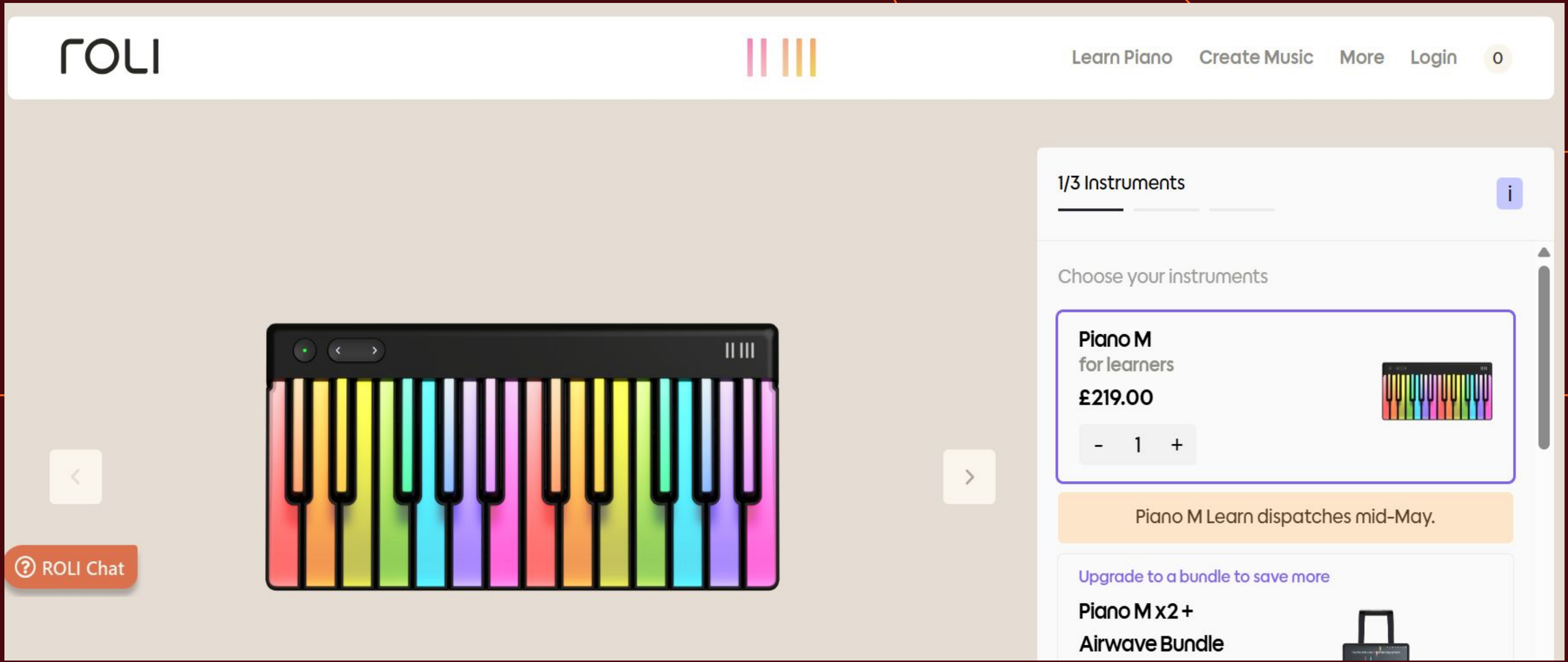
(1) https://musictechnology.uk/uk-music-tech-sector-faces-critical-lead-or-lose-moment-new-report-reveals/
(2) https://www.mordorintelligence.com/industry-reports/online-music-education-market
(3) https://assets.publishing.service.gov.uk/media/685943ddb328f1ba50f3cf15/industrial_strategy_creative_industries_sector_plan.pdf
(4) https://mooreks.co.uk/insights/the-createch-report-2021-venture-capital-firms-back-uk-as-global-createch-hub/
(5) https://lordslibrary.parliament.uk/access-to-music-education-in-schools/



metric_group	step_count	mean_ms	max_ms	mean_abs_ms	max_abs_ms
AUTOPLAY_TIMING_ERROR	12	29.667	61	32.667	61
LED_CMD_LATENCY	12	2.25	3		
SERVO_CMD_LATENCY	12	0.667	1		

“For interaction in a networked music performance to feel natural, the latency generally must be kept below 30 milliseconds, the bound of human perception…” –Wikipedia

https://en.wikipedia.org/wiki/Networked_music_performance


roli
Learn Piano
Create Music
More
Login
0
1/3 Instruments
i
Choose your instruments
Piano M
for learners
£219.00
- 1 +
Piano M Learn dispatches mid-May.
Upgrade to a bundle to save more
Piano M x2 +
Airwave Bundle
ROLI Chat